Learn, Denoise and Discover:

AI-powered Denoising for Electron Microscopy

Motivation: Studying catalysis

90% of all manufactured goods involve catalytic processes somewhere in their production chain

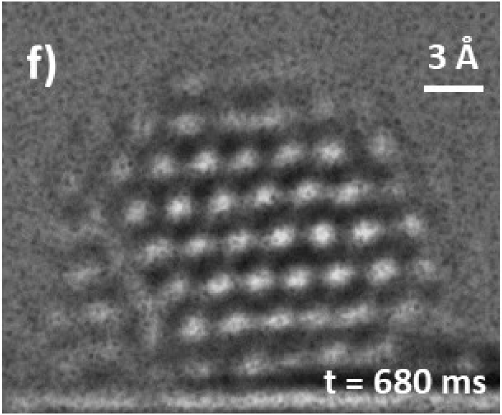
Important for energy, pharmaceuticals, materials, transport, and environmental applications

Nanoparticle-based catalysts are crucial

To understand functionality, we need to observe changes in the atomic structure of nanoparticles

Challenge: Atoms are tiny!

Order of magnitude: $1\ \text{Å} = 10^{-10}$ m

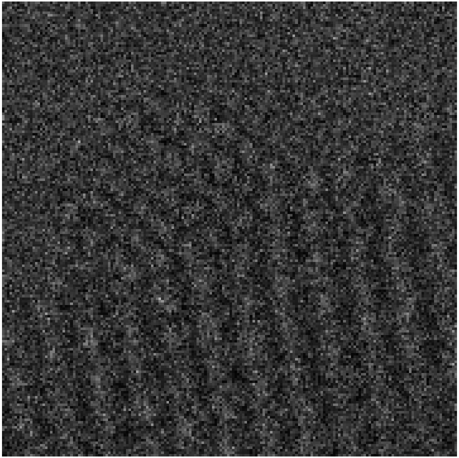


Electron microscopy enables visualization of the atomic structure by shooting electrons at the material

Challenge: Atomic structure changes fast!

We need to acquire data as fast as possible

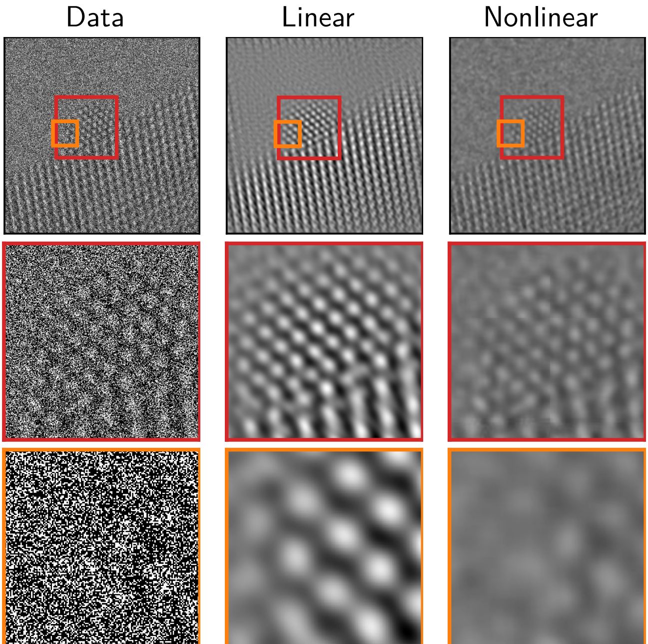
Acquisition at a timescale of milliseconds $\Longrightarrow$ Weak signal



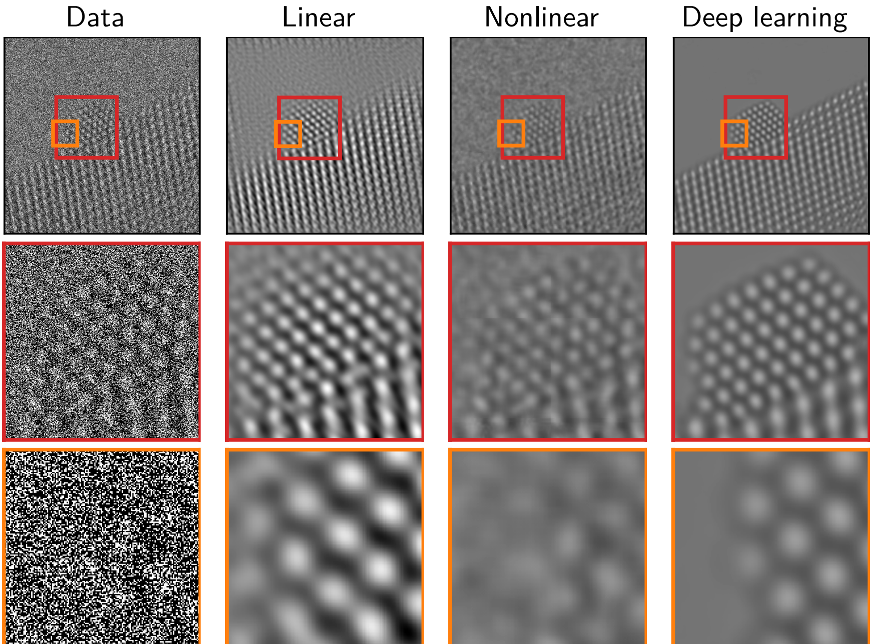
Possible solution: Higher electron dose $\Longrightarrow$ **destroys material**

We need to denoise!

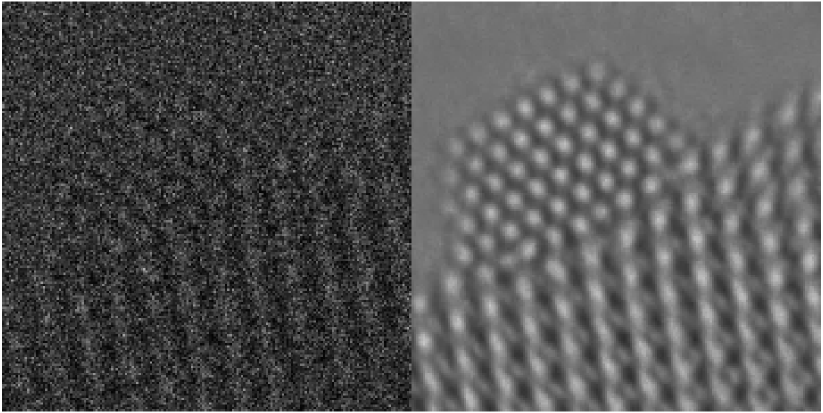
Traditional methods



AI-powered denoising



AI-powered denoising

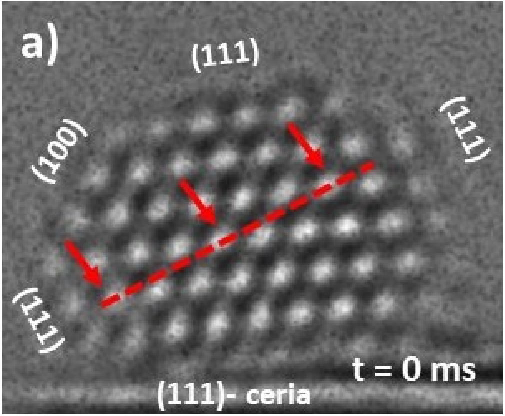


AI-powered denoising reveals unobserved dynamics

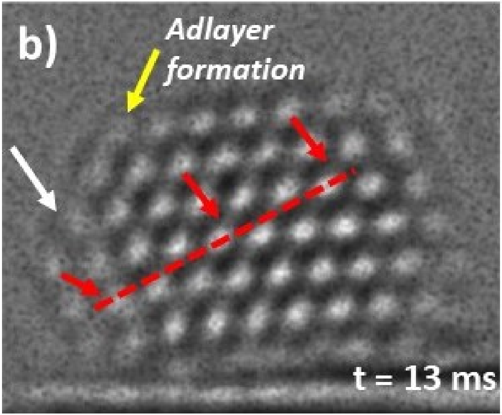
Visualizing nanoparticle surface dynamics and instabilities enabled by deep denoising. **Science** 387 (6737) pp:949-954

Crozier PA, Leibovich M, Haluai P, Tan M, Thomas AM, Vincent J, Mohan S, Marcos Morales A, Kulkarni SA, Matteson DS, Wang Y, Fernandez-Granda C

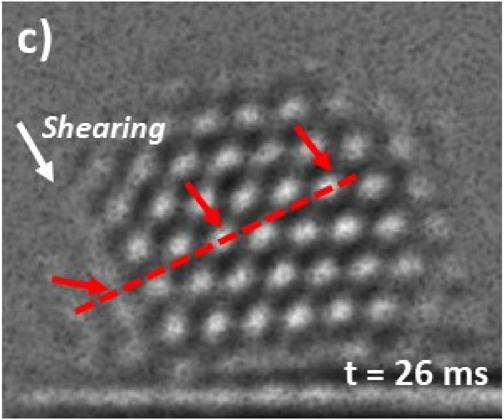
Shearing (displacement of atomic plane)



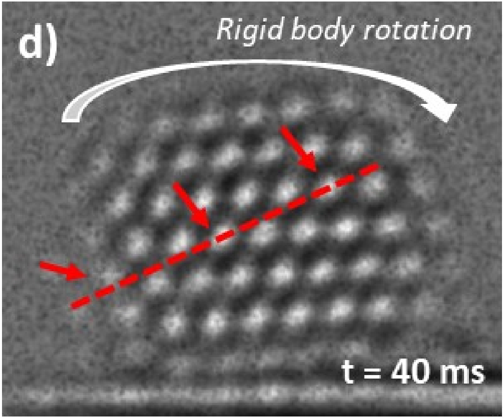
Shearing (displacement of atomic plane)



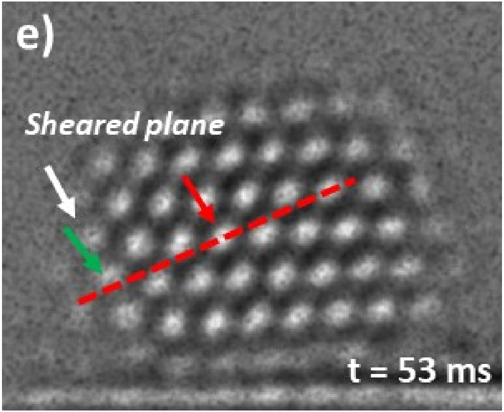
Shearing (displacement of atomic plane)



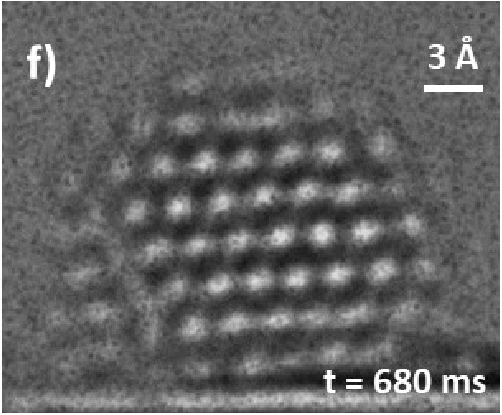
Rigid rotation



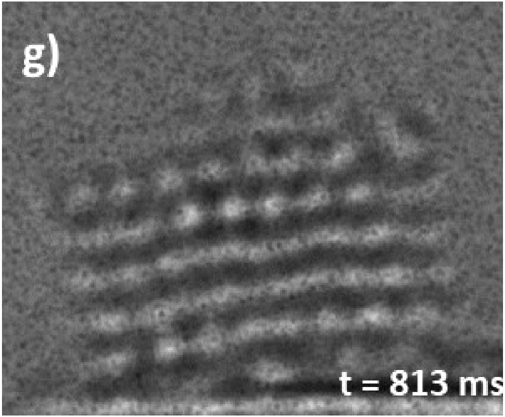
Rigid rotation



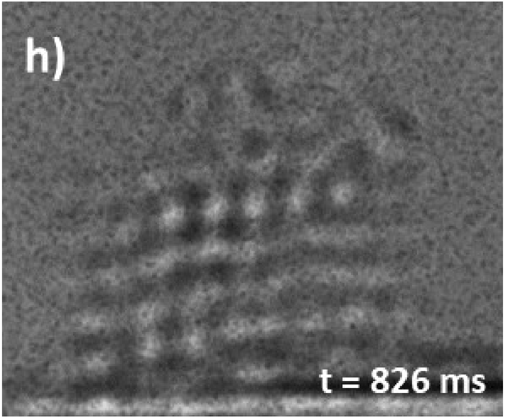
Destabilization (fluxionality)



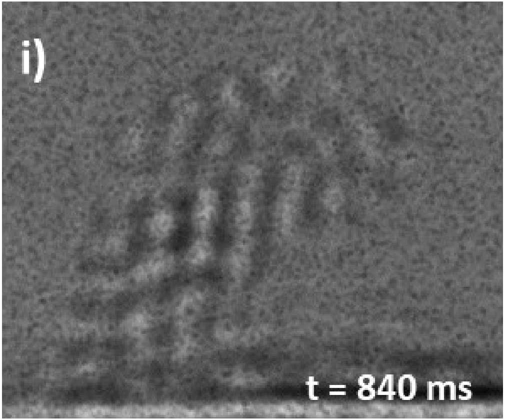
Destabilization (fluxionality)



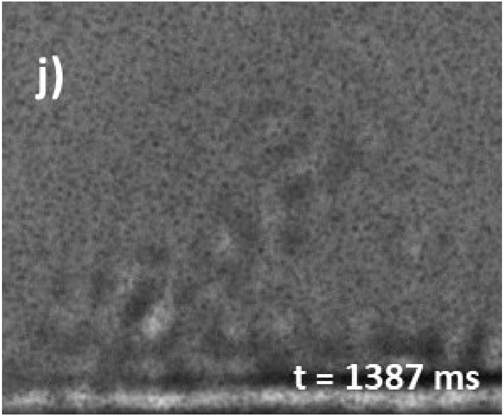
Destabilization (fluxionality)



Destabilization (fluxionality)



Destabilization (fluxionality)



Plan

Supervised learning

Unsupervised learning

Semi-supervised learning

Evaluation

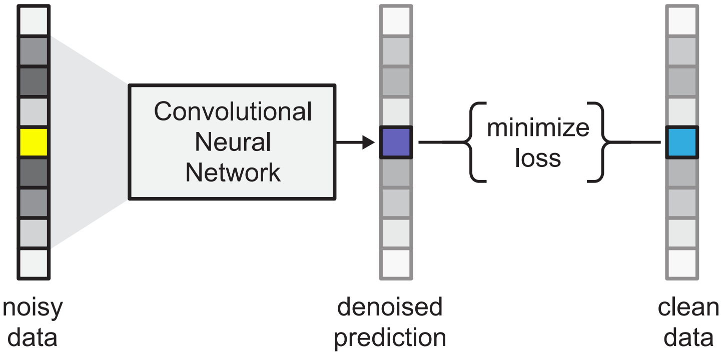
Supervised learning

Unsupervised learning

Semi-supervised learning

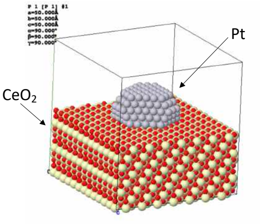
Evaluation

Supervised learning



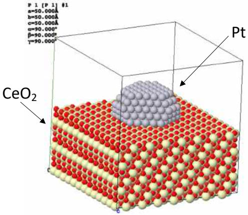
Problem: Requires access to clean - noisy pairs

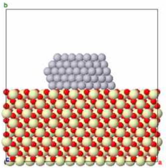
Simulation-based denoising



Simulating

Simulation-based denoising

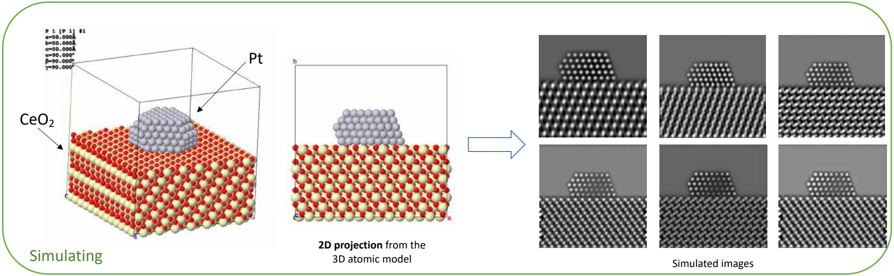




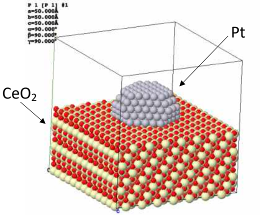
2D projection from the
3D atomic model

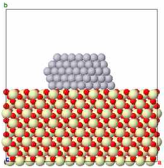
Simulating

Simulation-based denoising

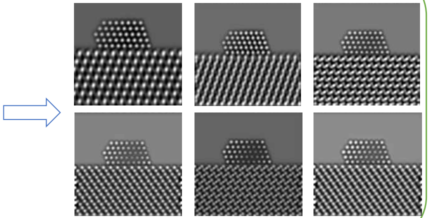


Simulation-based denoising



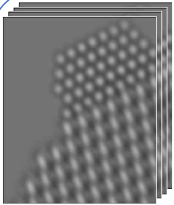


2D projection from the
3D atomic model



Simulated images

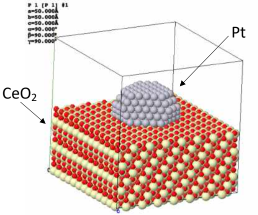
Simulating

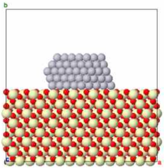


Simulated data

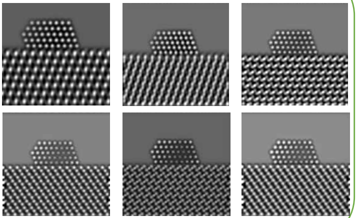
Training

Simulation-based denoising



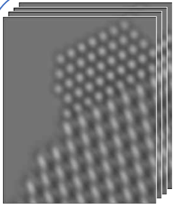


2D projection from the 3D atomic model



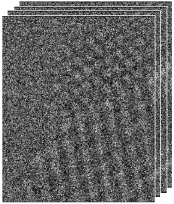
Simulated images

Simulating



Simulated data

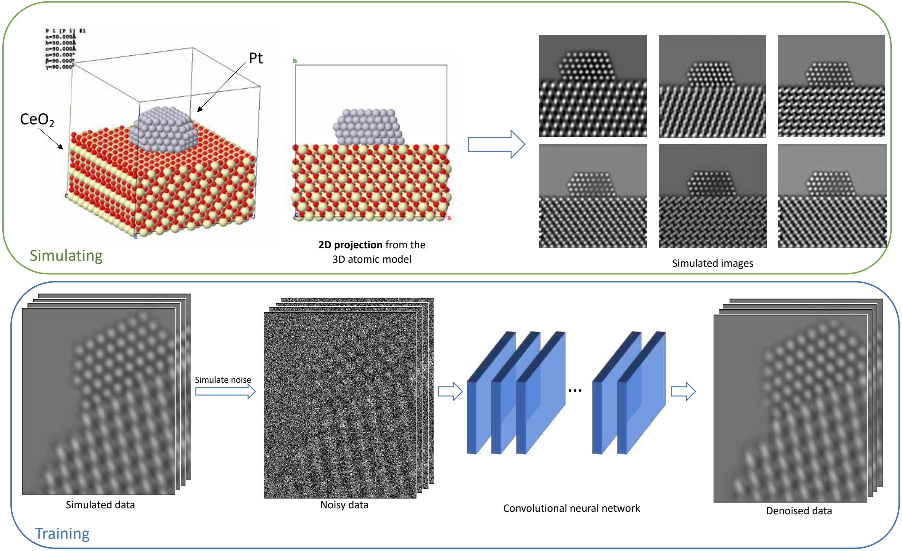
Simulate noise



Noisy data

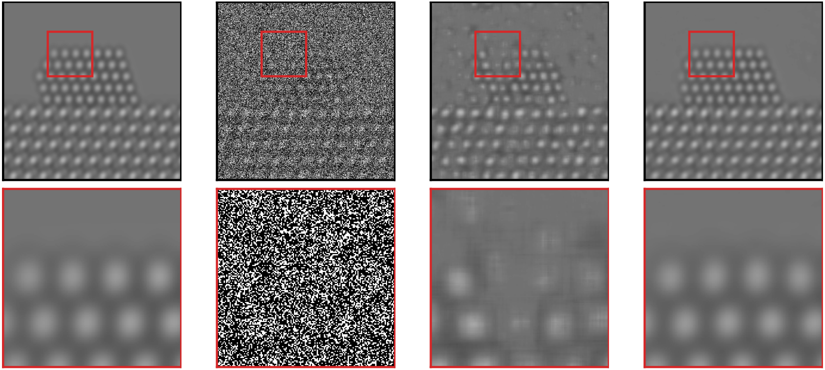
Training

Simulation-based denoising



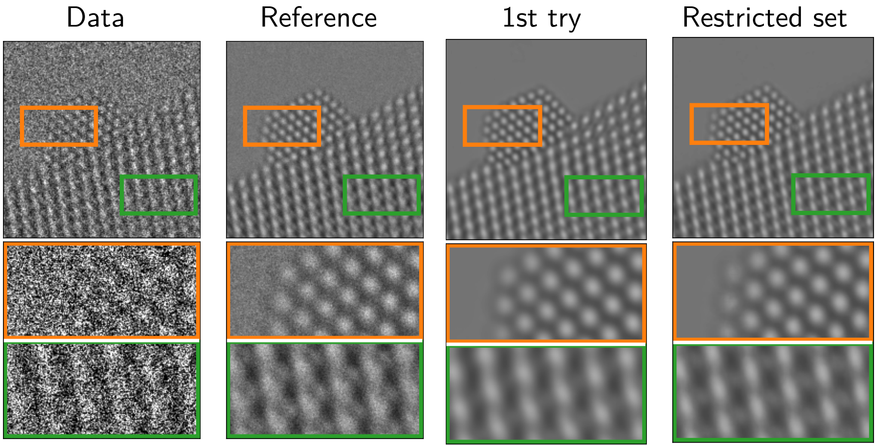
Deep denoising for scientific discovery: A case study in electron

Lesson 1: Field of view



Simulated clean image | Noisy image | DnCNN | UNet

Lesson 2: Design of training set



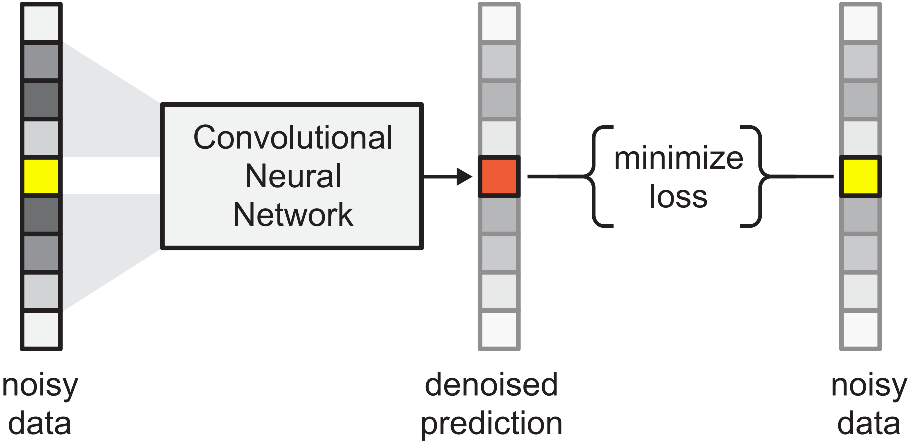
Supervised learning

Unsupervised learning

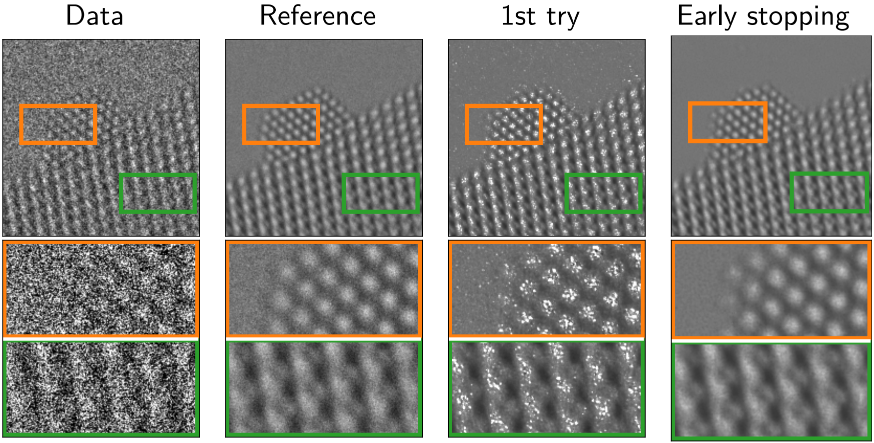
Semi-supervised learning

Evaluation

Blindspot denoising



Lesson: Unsupervised validation is crucial



Unsupervised Deep Video Denoising. Sheth et al. 2021

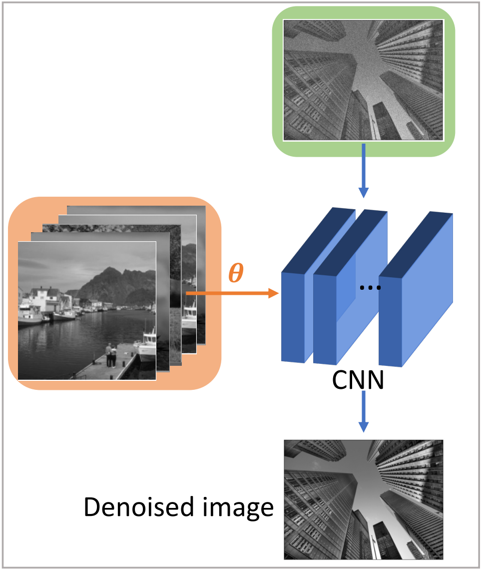
Supervised learning

Unsupervised learning

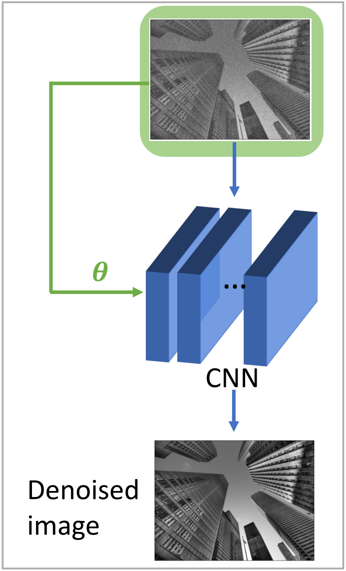
Semi-supervised learning

Evaluation

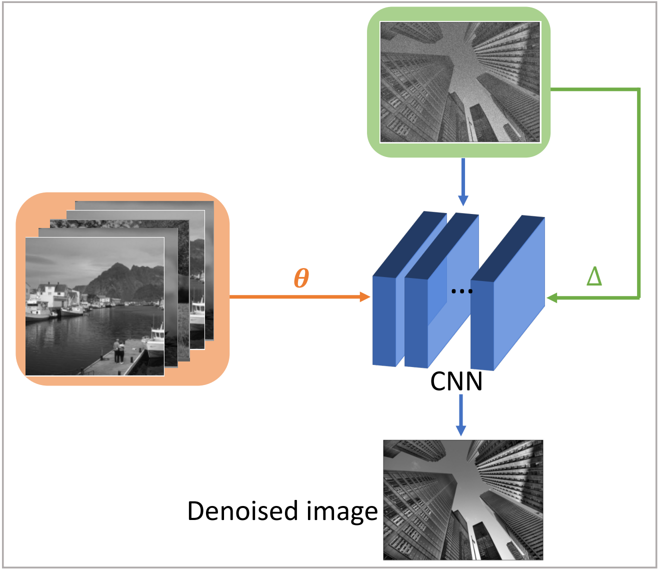
Supervised learning



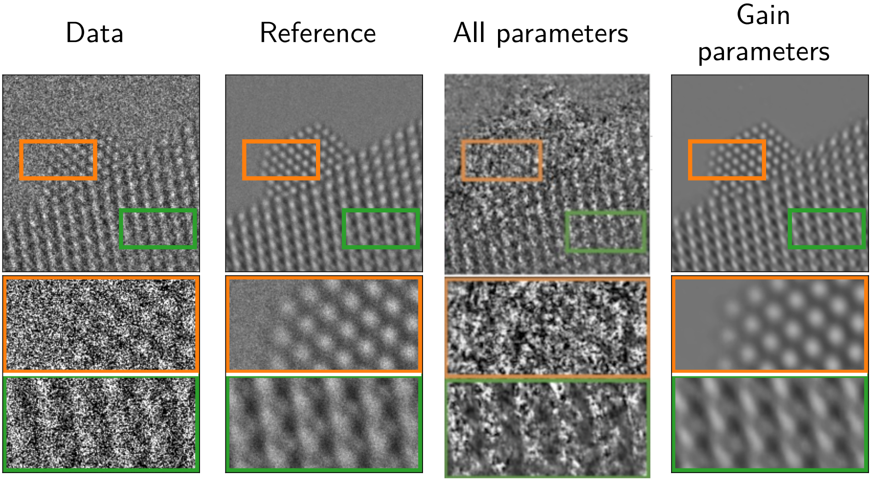
Unsupervised learning



Semi-supervised learning



Lesson: Cannot finetune all parameters



Adaptive denoising via gaintuning. Mohan et al. 2021

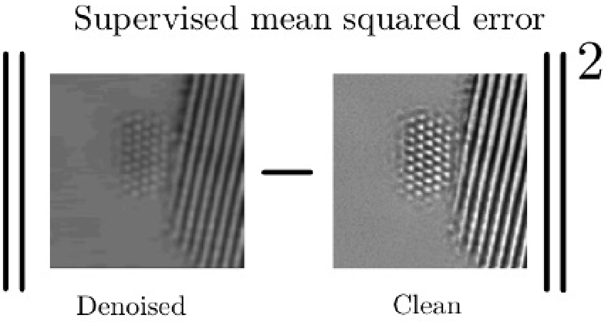
Supervised learning

Unsupervised learning

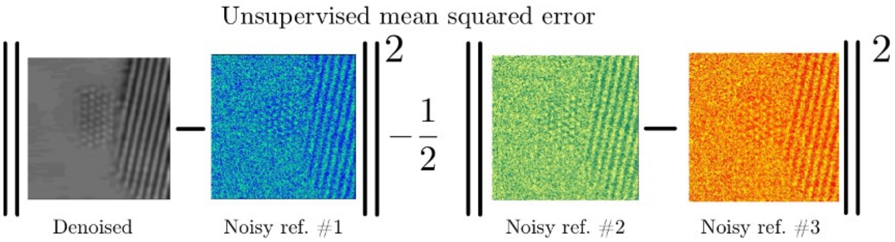
Semi-supervised learning

Evaluation

Supervised evaluation is not possible



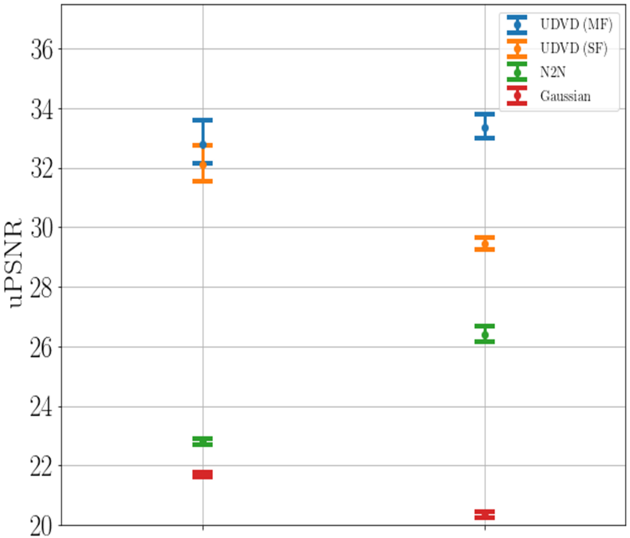
Unsupervised evaluation



Unbiased consistent estimator of MSE (assuming noise is independent across frames)

Evaluating unsupervised denoising requires unsupervised metrics. Marcos-Morales et al. 2023

Unsupervised evaluation



Beyond MSE

MSE may not capture scientifically-meaningful structure

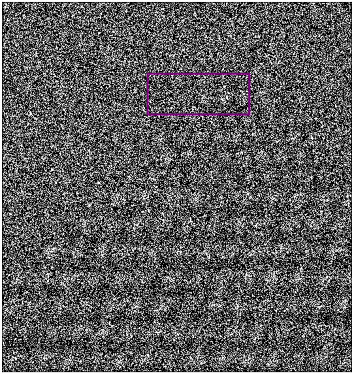
For example, presence or absence of atoms

Idea: Estimate *likelihood* of observed data given detected atoms

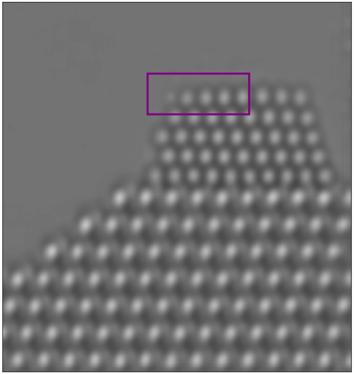
Deep denoising for scientific discovery: A case study in electron microscopy. Mohan el al. 2022

Beyond MSE

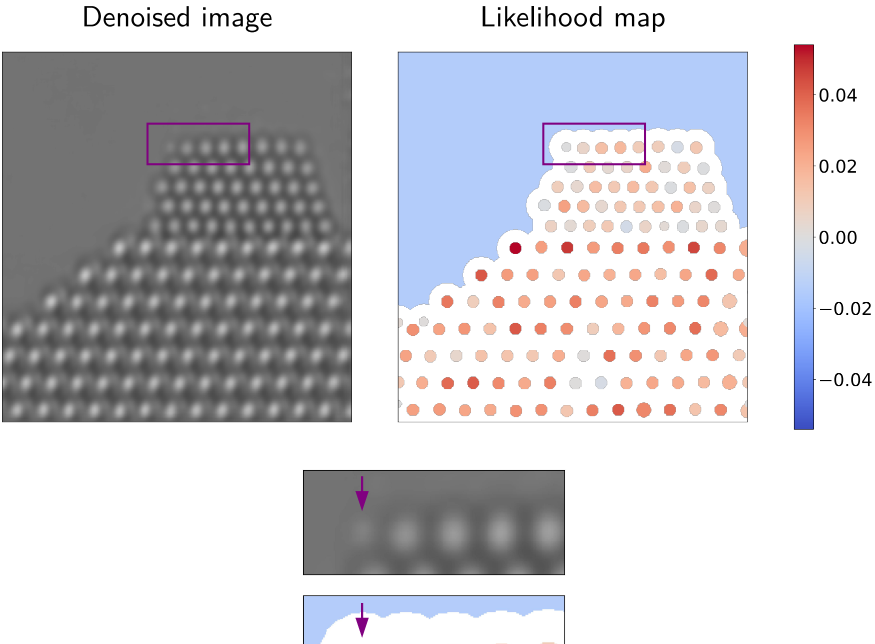
Data



Denoised image



Likelihood map



Conclusion

AI-powered denoising reveals hidden structure in electron microscopy data

Supervised, unsupervised and semi-supervised methods work, but overfitting is always looming!

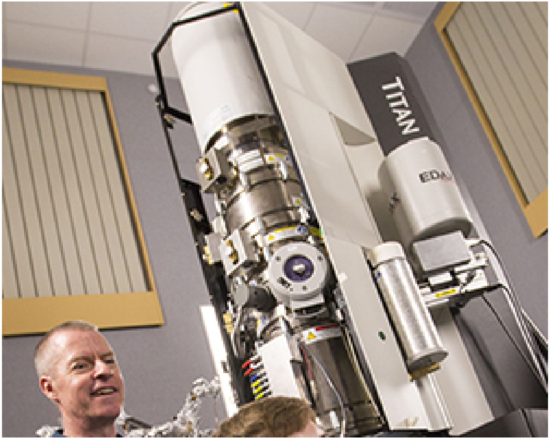
Evaluation is a major challenge

For more information

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan, Kangning Liu, Peter Crozier,
Carlos Fernandez-Granda

For more information



Crozier lab at Arizona State University

Acknowledgements

We are very grateful for the support of the National Science Foundation through grants OAC-2104105 and OAC-2103936